MindGlow – AI-Powered Youth Wellness Platform

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1 Project Overview

MindGlow is a web-based platform designed to support the mental wellbeing of young people by helping them monitor and better understand factors that influence stress and anxiety. The project was inspired by the findings and themes of the HBSC (Health Behaviour in School-aged Children) study, particularly those related to mental health, stress, sleep habits, emotional wellbeing, and healthy lifestyles.

The platform combines wellness tracking, data visualization, personalized recommendations, daily challenges, and an AI-powered wellbeing assistant into a single interactive solution.

2 Problem Statement

Stress and anxiety have become increasingly common among young people due to school pressure, social expectations, excessive screen time, poor sleep habits, and other lifestyle factors. Many adolescents are unable to recognize patterns in their behavior that negatively affect their wellbeing or lack accessible tools that encourage healthy habits and self-reflection.

MindGlow addresses this problem by providing a digital environment where users can track their wellbeing, visualize trends, receive personalized guidance, and communicate with an AI assistant that uses their wellness data to provide more relevant support.

3 Target Audience

The platform is primarily intended for adolescents and young people who experience stress and anxiety, school-related pressure, sleep difficulties, poor wellbeing habits, and challenges related to mental wellbeing.

4 Main Features

Daily Wellness Tracking. Users can submit daily wellness entries containing information such as mood level, stress level, anxiety level, sleep duration and quality, physical activity, screen time, social interaction, school pressure, and personal notes.

Wellness Dashboard. The platform visualizes collected data through interactive charts and statistics, allowing users to identify trends and better understand the relationship between their habits and emotional wellbeing.

Wellness Score. Based on the collected data, the system calculates a wellness score that provides users with a simple indicator of their overall wellbeing for a given day.

Daily Challenges. MindGlow encourages positive behavior through daily wellness challenges designed to promote healthier habits, emotional awareness, physical activity, and stress reduction.

Personalized Recommendations. The system analyzes recent wellness data and generates recommendations tailored to the user's current situation and needs.

AI Wellbeing Assistant. One of the core features of the platform is an AI-powered assistant that can discuss stress and anxiety-related topics, provide supportive and empathetic responses, suggest healthy coping strategies, and offer personalized advice based on the user's wellness history.

The assistant is intended as a supportive wellbeing tool and is not presented as a replacement for professional psychological or medical help.

5 Technologies Used

Frontend. The frontend is implemented using React, Tailwind CSS, and Recharts for data visualization.

Backend. The backend is implemented using Django and Django REST Framework.

Database. The system uses PostgreSQL as the main database.

Artificial Intelligence. The AI assistant is powered by the OpenAI API.

6 Expected Impact

The goal of MindGlow is to encourage self-awareness and healthy habits among young people. It helps users recognize patterns related to stress and anxiety, improve sleep and daily routines, develop healthier digital habits, build emotional literacy, engage in positive daily activities, and reflect on their wellbeing in a structured manner.

Through continuous tracking, personalized recommendations, and AI-assisted support, the platform aims to contribute to improved mental wellbeing and healthier lifestyles among young people.

7 Project Repository

GitHub Repository:

`https://github.com/Itonkdong/mindglow`

8 Live Demonstration

Application URL:

https://mindglow-frontend.onrender.com

Note: The application is hosted on Render's free tier. If the service is inactive, the initial load may take approximately one minute while the instance starts.